

Supporting material for Myopic Sampling Can Be Maximally Suboptimal in Risk-Limiting Financial Audits

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This supplement collects the support correction, implementation specialization, conditional interval-policy results, and numerical material accompanying the short note. The note's sharp factor- N theorem and cost corollary have self-contained proofs and do not depend on this material.

S1 Notation and the sharp family

Let $\pi_i > 0$ sum to one, $f_i \in [0, 1]$, and $m^\star = \sum_i \pi_i f_i$. With R_t the remaining set and A_{t-1} the observed contribution, use strictly positive sampling probabilities and the wealth update

$$Z_t = \frac{\pi_{I_t} f_{I_t}}{q_t(I_t)}, \quad W_t(m) = W_{t-1}(m) \{1 + \lambda_t(m)[Z_t - (m - A_{t-1})]\}, \quad W_0(m) = 1. \quad (\text{S1})$$

Stakes are predictable, valid (the wealth factors are nonnegative), and uniformly bounded by L . The confidence set is the running intersection of $\{m \in [0, 1] : W_t(m) < 1/\delta\}$ and the logical intervals

$$J_t = \left[A_t, A_t + \sum_{i \in R_{t+1}} \pi_i \right]. \quad (\text{S2})$$

The stopping time is the first draw with diameter at most ε ; $\text{diam } \emptyset = 0$. The oracle is

$$q_t^{\text{or}}(i) = \frac{\pi_i f_i}{\sum_{j \in R_t} \pi_j f_j}. \quad (\text{S3})$$

The note's construction chooses rational $\varepsilon > 0$ with

$$\varepsilon < 1/3, \quad (1 + 2L\varepsilon)^N < 1/\delta, \quad (\text{S4})$$

and rational $\rho \in (0, 1]$, and sets

$$a = \varepsilon/2, \quad w = a/(N - 1), \quad (\text{S5})$$

$$\pi_0 = 1 - a, \quad f_0 = \frac{\rho w}{1 - a}; \quad \pi_i = w, \quad f_i = 1 \quad (1 \leq i < N). \quad (\text{S6})$$

The two candidates used in the proof are

$$m_- = a, \quad m_+ = 5a. \quad (\text{S7})$$

The main theorem gives

$$V^* = 1, \quad \mathbb{E}\tau_{\text{or}} = 1 + \frac{N-1}{1+\rho}, \quad (\text{S8})$$

where V^* is an infimum over fully supported policies for the fixed betting rule. Giving item 0 cost one and each other item cost $\kappa \geq 1$ gives

$$V_c^* = 1, \quad \mathbb{E}C_{\text{or}} = 1 + \frac{(N-1)\kappa}{1+\rho}. \quad (\text{S9})$$

Neither value statement asserts general nonattainment.

S2 Strict support does not imply nonattainment

An earlier draft incorrectly described both infima in (S8) and (S9) as always unattained under strict support. The sharp bounds require only the values of these infima. The following example shows why nonattainment must not be claimed for the full bounded-betting class.

Take $N = 2$, $\delta = 1/20$, $L = 1$, $\varepsilon = 1/40$, and

$$\pi = (79/80, 1/80), \quad f = (1/79, 1), \quad q_1 = (1679/1680, 1/1680).$$

These are the sharp-family parameters $\rho = 1$, $a = w = 1/80$ and satisfy $(1 + 2L\varepsilon)^2 = (21/20)^2 < 20$. Use

$$\lambda_1(m) = \mathbf{1}\{m > \varepsilon\}, \quad \lambda_t(m) = 0 \quad (t \geq 2).$$

This is a predictable rule using the known tolerance. It is valid for every possible taint vector: $Z \geq 0$ and $m \leq 1$ imply $1 + Z - m \geq 0$ whenever the stake is one, while a zero stake gives factor one. Its uniform cap is one.

If the large item is drawn first, the logical interval is $[1/80, 1/40]$. All its candidates have stake zero, so the combined set is exactly this interval. If the small item is drawn first, $Z_1 = 21$ and $J_1 = [1/80, 1]$. For every $m > \varepsilon$, the wealth is $22 - m \geq 21 > 20$, so that candidate is rejected. All $m \leq \varepsilon$ retain wealth one. The combined set is again $[1/80, 1/40]$.

Thus the fully supported policy has $\tau = 1$ on both branches, and the true total $m^* = 1/40$ remains covered on both. Unit costs also give attained expected cost one. The separate zero-first-stake result below retains its more restrictive nonattainment conclusion.

S3 The exact two-transaction problem

Assume throughout this section that the first stake is zero. This makes the first-round betting set all of $[0, 1]$. After sampling item i , the combined interval therefore has diameter $1 - \pi_i$. Define

$$A(\pi, \varepsilon) = \{i \in \{1, 2\} : 1 - \pi_i \leq \varepsilon\}.$$

Proposition S1 (Complete $N = 2$ solution). *For every fully supported first action q ,*

$$\mathbb{E}_q[\tau] = 2 - \sum_{i \in A(\pi, \varepsilon)} q(i). \quad (\text{S10})$$

Hence the full-support infimum is 2 if $\varepsilon < \min\{\pi_1, \pi_2\}$, and 1 otherwise. When A is a singleton, the infimum is unattained; when A is empty or contains both items, every fully supported first action is optimal.

Proof. The first draw stops exactly on A . Otherwise the sole remaining item is audited at time two and (S2) is a singleton. Thus $\mathbb{E}_q[\tau] = q(A) + 2\{1 - q(A)\}$, and optimizing gives the claim. $\square$

For

$$(\pi_1, \pi_2) = \left(\frac{3}{4}, \frac{1}{4}\right), \quad (f_1, f_2) = \left(\frac{1}{3}, 1\right), \quad \varepsilon = \frac{1}{3}, \quad \delta = \frac{1}{20},$$

the full-support infimum, prop-M, and oracle expectations are respectively

$$1 < \frac{5}{4} < \frac{3}{2}. \quad (\text{S11})$$

Under strict full support the infimum remains one but is not attained; putting probability $1 - \eta$ on item 1 gives expectation $1 + \eta$.

S4 ApproxKelly specialization

Set $\delta = 1/20$, $L = 5/2$, and $\varepsilon = 1/(20N)$. Then

$$(1 + 2L\varepsilon)^N = \left(1 + \frac{1}{4N}\right)^N < e^{1/4} < 2 < 20.$$

The two witnesses in (S7) also lie at indices 1 and 5 of the allowed uniform grid with $40N + 1$ points. Thus the released ApproxKelly construction is a corollary, but neither its zero initialization nor its grid drives the general theorem.

Prop-M stops no later than the rank of item 0, whose expectation is

$$1 + \frac{(N-1)w}{\pi_0 + w} = 1 + \frac{a}{\pi_0 + w} < \frac{1}{\pi_0}, \quad (\text{S12})$$

which approaches one. If point scores happen to equal the realized taints, $S_i = f_i$, prop-MS is identical to (S3) and inherits both sharp lower bounds. This is a statement about the constructed scores and outcomes, not a claim that the point model has a certified zero-error guarantee.

The mathematical update uses the exact denominator in (S1). The released code adds a floating-point stabilizer; no claim of bit-for-bit numerical equivalence for arbitrarily small probabilities is made.

S5 Certified AI intervals: an exact minimax policy

The negative theorems say that point contributions do not encode terminal uncertainty. Suppose instead that a model, prior audit, or calibrated procedure supplies simultaneous bounds, clipped to $[0, 1]$, with

$$0 \leq \ell_i \leq f_i \leq u_i \leq 1 \quad (1 \leq i \leq N). \quad (\text{S13})$$

After auditing a set B , the exact image of this box under the total is

$$C_B^{\text{box}} = \left[\sum_{i \in B} \pi_i f_i + \sum_{i \notin B} \pi_i \ell_i, \sum_{i \in B} \pi_i f_i + \sum_{i \notin B} \pi_i u_i \right]. \quad (\text{S14})$$

Both endpoints are attainable because the uncertainty set is a Cartesian product. Define the dollar-weighted uncertainty

$$d_i = \pi_i(u_i - \ell_i), \quad D(B) = \sum_{i \notin B} d_i. \quad (\text{S15})$$

The observation changes the location of (S14), but not its width $D(B)$.

This section uses a separate policy class: items may be selected deterministically, and $\tau_{\text{box}} = \min\{0 \leq t \leq N : D(B_t) \leq \varepsilon\}$, where B_t is the set reviewed after t draws. No importance weighting is used. The strict-support restriction from Section S1 therefore does not apply to the box-only certificate.

Theorem S1 (Pathwise minimax box policy). *Let $d_{(1)} \geq \dots \geq d_{(N)}$, and define*

$$k^* = \min \left\{ k \in \{0, \dots, N\} : \sum_{r=k+1}^N d_{(r)} \leq \varepsilon \right\}. \quad (\text{S16})$$

Auditing in nonincreasing order of d_i stops (S14) in exactly k^ reviews. Moreover, over all adaptive randomized policies,*

$$\inf_q \sup_{f \in \Pi_i[\ell_i, u_i]} \mathbb{E}_f[\tau_{\text{box}}(q)] = k^*. \quad (\text{S17})$$

The lower bound holds on every sample path.

Proof. The sorted policy leaves the residual in (S16). Any history of length $t < k^*$ can remove at most $\sum_{r=1}^t d_{(r)}$. Its residual width therefore exceeds ε , regardless of observations, randomization, or adaptation. No policy can stop earlier on any path, while the sorted construction attains the bound. $\square$

This gives a precise answer to the implementable robust-certificate problem: audit uncertainty, not merely predicted error. For a multiplicative guarantee, write the condition without division as

$$(1 - b)f_i \leq S_i \leq (1 + b)f_i, \quad 0 \leq b < 1. \quad (\text{S18})$$

It implies

$$\ell_i = \frac{S_i}{1 + b}, \quad u_i = \min \left\{ 1, \frac{S_i}{1 - b} \right\}. \quad (\text{S19})$$

If no upper endpoint clips, then

$$d_i = \pi_i S_i \frac{2b}{1 - b^2}. \quad (\text{S20})$$

Thus $\pi_i S_i$ is the correct priority, but it should be used as a descending deterministic priority for this certificate, not as a randomized probability. When (S19) clips, the exact priority in (S15) can differ even in rank from $\pi_i S_i$.

Corollary S1 (Point accuracy versus certified audit value). *On the population (S5)–(S6), let the point scores happen to equal the realized taints, $S_i = f_i$, and supply the simultaneous intervals*

$$[\ell_0, u_0] = \left[0, \frac{2\varepsilon}{\pi_0} \right], \quad [\ell_i, u_i] = [0, 1] \quad (1 \leq i < N). \quad (\text{S21})$$

The certified policy stops in one review. Randomized prop-MS has

$$\mathbb{E}[\tau_{\text{propMS}}] = 1 + \frac{N - 1}{1 + \rho}.$$

With $c_0 = 1$ and $c_i = \kappa$, its expected review cost is (S9), while the certified cost optimum is one.

Proof. The large upper endpoint is below one because $\varepsilon < 1/3$. The intervals contain the realized taints: in particular, $\pi_0 f_0 = \rho w \leq 2\varepsilon$. Their uncertainty contributions satisfy

$$d_0 = 2\varepsilon, \quad \sum_{i=1}^{N-1} d_i = \sum_{i=1}^{N-1} w = a = \frac{\varepsilon}{2}.$$

No audited set omitting item 0 can stop, because its residual contains $d_0 > \varepsilon$. Auditing item 0 leaves residual $a \leq \varepsilon$, so descending uncertainty stops after one review. For the box stopping rule, $S = f$ makes prop-MS use the oracle rates, and it stops exactly at item 0's rank. The review and cost formulas follow from the exponential-clock calculation. $\square$

S5.1 Heterogeneous costs

For general positive c_i , the exact robust problem becomes

$$\begin{aligned} \min_{x \in \{0,1\}^N} \quad & \sum_i c_i x_i \\ \text{s.t.} \quad & \sum_i d_i x_i \geq D(\emptyset) - \varepsilon. \end{aligned} \tag{S22}$$

The repository implements an exact rational Pareto-frontier dynamic program. After each item, a state records removed width and cost, caps width at the target, and discards a state if another has at least as much removed width at no greater cost. Induction over items proves that every discarded partial solution is dominated and that the least-cost terminal state solves (S22).

S5.2 Risk accounting and a hybrid audit

Suppose (S13) holds simultaneously with probability at least $1 - \delta_{\text{AI}}$. On that event, (S14) covers m^* at every time, under any adaptive policy. If a betting CS has simultaneous failure probability δ_{CS} , their running intersection fails with probability at most

$$\delta_{\text{AI}} + \delta_{\text{CS}} \tag{S23}$$

by a union bound; independence is unnecessary. A deployable architecture can therefore individually examine a certainty stratum chosen by (S15), then run a randomized risk-limiting procedure on the residual population, explicitly allocating the two risk budgets.

S6 Finite-grid Bellman verification

For general N , the state must contain the ordered history, remaining set, wealth at every candidate m , ApproxKelly payoff accumulators, and current running intersection. If $\Phi(x, q, i)$ is the transition, the exact equation is

$$V(x) = \begin{cases} 0, & x \text{ is in the stopping region,} \\ 1 + \inf_q \sum_{i \in R(x)} q(i) V(\Phi(x, q, i)), & \text{otherwise.} \end{cases} \tag{S24}$$

The action changes both transition probability and confidence state. This is the continuation value absent from the one-step surrogate.

Table S1: Expected stopping times on the finite grid/mesh, rounded from exact rationals. All policies use oracle payoff bounds. Mesh priority is randomized.

Case	Oracle	prop-M	prop-MS	Mesh priority	Mesh Bellman
Terminal value, $N = 3$	2.4167	2.1500	2.3158	2.0556	2.0556
Mixed information, $N = 4$	3.8558	3.4488	3.8302	3.7176	3.4213
Clipped score, $N = 4$	3.2008	3.4488	3.1839	3.4213	3.4213

Table S2: Synthetic certified-box workload; not a client-data claim.

Scenario	Sorted optimum	prop-MS	Oracle	prop-M	Uniform
Unclipped, concentrated	347	452.13	452.68	493.41	895.66
Clipped, high risk	412	536.52	531.01	528.11	898.89
Diffuse values	740	819.31	819.41	895.66	900.44
Tight certificate	659	736.39	736.64	770.81	975.57

We implement (S24) with exact rational arithmetic on a finite candidate grid and the strict-support mesh $q_i \in \{1/D, \dots, D/D\}$. Backward induction over the finite horizon gives the global optimum over that finite action set; no simulation enters these values. Table S1 uses a 21-point candidate grid and $D = 6$. Every policy’s admissible betting range uses the unrevealed true taints (`payoff_information="oracle"`). These are fixed-population comparisons with oracle information, not evaluations of implementable betting policies. The mesh-priority comparator randomizes on the strict-support mesh according to certified-width priorities; it is not the deterministic optimal box policy. Policies not lying on this action mesh can beat its optimum, so the table is verification and structure discovery, not a continuous-action optimality claim.

The first mesh action in the mixed $N = 4$ case is $(1/6, 1/6, 1/2, 1/6)$, concentrating on neither the largest recorded value nor the largest contribution alone. This is numerical evidence for a blended continuation-value index, not yet a theorem.

S7 Reproducible stress tests

We generated four deterministic $N = 1000$ synthetic populations from fixed seeds. Each AI interval follows (S19); the stopping rule is solely the simultaneous-box width. The sorted optimum is exact. Comparator means use 1,000 independent exponential-race permutations, which exactly generate their probability-proportional-without-replacement orders. Table S2 reports mean reviews; Monte Carlo standard errors for prop-MS range from 0.17 to 0.30 reviews.

The exact policy uses 9.7–23.3% fewer reviews than randomized prop-MS in these scenarios. These are controlled stress tests of the theorem’s mechanism, not estimates of savings in any audit population. Real validation requires authorized data, simultaneous calibration checks, review-cost measurement, and comparison under the applicable audit objective.

S8 Verification artifacts

The repository [3] contains two exact rational JSON certificates, independent verification scripts, the complete two-item characterization, and tests for the arbitrary- N witness inequalities and cost formulas. The support-attainment example in Section S2 has a separate exact rational regression check. Certificates corroborate finite instances; they do not mechanically certify the general theorem.

The synthetic tables use simulated random sampling orders and are labeled as such. The finite-grid table instead uses exact rational backward induction on its explicitly stated oracle-information model. Neither table establishes field performance or solves the unrestricted continuous-action problem.

References

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